DA6701 – Assignment 2

Team 14

1 Introduction

The objective of this project is to design a predictive system for a six-stock Indian equity universe (Reliance (RELIANCE), HDFC Bank (HDFCBANK), Infosys (INFY), Mahindra (MM), Bharti Airtel (BHARTIARTL) and Hindustan Unilever (HUL)) that forecasts daily returns using multiple information sources.

The system integrates structured market data, firm fundamentals, macroeconomic indicators, and financial news sentiment to construct a unified machine learning pipeline for return prediction. Using these forecasts, a forward-looking portfolio allocation strategy must be developed and evaluated over an out-of-sample test period. The model must operate under realistic financial constraints, including temporal consistency, absence of look-ahead bias, and robustness across all securities.

2 Data Sources

The predictive framework requires us to integrate financial data like market prices, firm fundamentals, macroeconomic indicators, capital flows, commodities, currency rates, volatility indices, and textual sentiment signals.

Table 1: Summary of Data Sources and Features

Source	Columns	Frequency
Yahoo Finance	open, high, low, close, volume	Daily
Computed Features	stock_log_return, volume_change	Daily
Perplexity Finance	eps, pe_ratio, revenue, net_income, profit_margin, ...	Quarterly
Yahoo Finance	nifty50_return, sector_return	Daily
NSE India VIX	india_vix	Daily
Reserve Bank of India	repo_rate, tbill_91d, gsec_10y_yield, yield_curve_spread	Daily/Weekly
Reserve Bank of India	cpi, cg_securities_cr, sg_securities_cr	Monthly
Trendlyne	fii_net_cr, dii_net_cr	Monthly
Reserve Bank of India	usd_inr, gbp_inr, eur_inr, jpy_inr	Daily
Investing.com	crude_brent_return, gold_return	Daily
GDELT + FinBERT	sentiment_score	Daily

3 Methodology

3.1 Data Preprocessing

The following preprocessing procedures were applied to ensure data quality and temporal consistency:

- Features with more than 15% missing observations were removed to avoid unreliable signals and unstable model estimates.

- Remaining missing values were handled using forward filling where economically and temporally appropriate, ensuring that only information available at each timestamp was used.
- Log returns were computed from adjusted closing prices to obtain scale-invariant and approximately stationary series suitable for modeling.
- Rows containing missing or infinite values in required predictor or target columns were discarded, primarily corresponding to rolling-window warm-up periods.
- Extreme values were controlled via symmetric winsorisation at the 1% tails, reducing the influence of outliers while preserving distributional structure.
- Predictors were temporally shifted wherever required so that each feature strictly preceded the prediction horizon, thereby eliminating implicit forward information.

To rigorously prevent data leakage, all preprocessing transformations (including filtering, winsorisation thresholds, scaling, and feature construction parameters) were estimated exclusively on the training set and subsequently applied unchanged to the forward test period.

3.2 Feature Engineering

Raw OHLC price levels were excluded because they are non-stationary and scale-dependent, which may cause models to learn spurious relationships driven by price magnitude rather than underlying market dynamics.

Instead, we construct technical indicators derived solely from historical prices—such as moving averages, momentum oscillators, and volatility measures—which convert raw price series into normalized, time-relative signals. These representations more effectively capture trend structure, persistence, and conditional variability while remaining robust to scale effects.

All features were constructed strictly using information available up to time t to ensure temporal causality and eliminate look-ahead bias. Details of the engineered technical indicators are provided in Table A.2.

3.2.1 Interaction Features and Rolling Standardisation

Pairwise interaction features were generated among the most informative predictors selected via mutual information, allowing the model to capture nonlinear joint effects between signals. Only cross-feature interactions were retained to avoid redundant higher-order terms.

All predictors were then standardised using a 30-day rolling window with a one-period shift,

$$z_t = \frac{x_t - \mu_{t-30:t-1}}{\sigma_{t-30:t-1}},$$

so that scaling statistics were computed strictly from past observations, ensuring temporal causality. Features not suitable for scaling were excluded, and missing values introduced during rolling computation were forward-filled and zero-imputed where necessary. Rows were retained as long as target values were present.

3.3 Feature Selection

Feature selection is performed in two stages: (i) construct the candidate feature pool using one of two approaches, and (ii) apply a common multi-method selection pipeline (Domain MI,

Random Forest, RFE, SHAP) followed by majority voting. The difference between Approach 1 and Approach 2 lies in how features are partitioned into *excluded*, *always-kept*, and *feature-selection* groups.

3.3.1 Approach 1: Broad Numeric Pool with Fundamentals Added Back

In Approach 1, all numeric columns are first identified and filtered by removing:

- meta columns (e.g., date),
- target columns,
- leaky columns (e.g., OHLC prices),
- redundant columns.

Fundamental variables are kept separate during initial filtering. The feature-selection matrix is then formed by combining:

- numeric candidate features (excluding meta/target/leaky/redundant columns), and
- fundamental features from `FUNDAMENTAL_COLS` that are available in the stock dataframe.

Thus, engineered features and core fundamentals jointly compete in the feature selection stage.

Table 2: Approach 1: Selected Additional Features by Stock

Stock	Count	Additional Features
BHARTIARTL	29	MACD_signal, return_lag3, volume_lag3, usd_inr, gbp_inr, eps, revenue, gsec_10y_yield, EMA_12, SMA_20
HDFCBANK	31	MACD_hist, MACD_signal, cg_securities_cr, tbill_91d, volume_z, revenue, profit_margin, volume, EMA_26, dii_net_cr, gbp_inr, usd_inr
HUL	30	volume_z, return_lag3, volume_lag3, usd_inr, cg_securities_cr, revenue, eps, EMA_12, eur_inr, SMA_20, volume
INFY	29	volume, gbp_inr, volume_z, sg_securities_cr, MACD_signal, repo_rate, MACD_hist, return_lag3, volume_lag3, revenue
MM	32	volume, gbp_inr, volume_lag3, return_lag3, revenue, net_income, MACD_hist, EMA_26, tbill_91d, cg_securities_cr, sg_securities_cr, profit_growth, SMA_20
RELIANCE	32	volume_lag3, volume, SMA_20, cg_securities_cr, MACD_signal, return_lag3, profit_margin, MACD_hist, EMA_26, eur_inr, usd_inr, tbill_91d, gsec_10y_yield

Common features across all stocks (19): ATR_14, BB_lower, BB_pct, BB_upper, Buying_Pressure, MACD, OBV, RSI_14, SMA_50, return_lag1, sector_lag1, volume_lag1, volume_change, india_vix, jpy_inr, gold_return, crude_brent_return, pe_ratio, sentiment_score.

3.3.2 Approach 2: Curated Always-Keep Variables + Technical FS Pool

Approach 2 uses a more structured design with three groups:

1. **Excluded columns** (`EXCLUDE`): leaky and redundant variables that are never used.
2. **Always-keep columns** (`INCLUDED_COLS`): variables retained directly in the final dataset.
3. **Feature-selection pool** (`candidate_features`): remaining numeric columns not in the above sets.

The always-keep group includes fundamentals, macro variables, flow variables, forex, commodity variables, selected volume features, and sentiment placeholders (if present), along with meta-data/targets for bookkeeping. As a result, feature selection is applied mainly to engineered technical and lag-based features (e.g., SMA/EMA, MACD, RSI, Bollinger-band features, ATR, OBV, and return/volume/sector lags), where redundancy is typically highest.

Table 3: Approach 2: Selected Additional Features by Stock

Stock	Count	Additional Features
BHARTIARTL	44	MACD_hist, volume_lag1, BB_lower, MACD_signal, BB_upper, BB_pct, MACD
HDFCBANK	46	volume_lag1, BB_upper, BB_lower, MACD_hist, MACD, MACD_signal, EMA_26, volume_z, SMA_50
HUL	45	BB_lower, MACD_hist, volume_z, BB_pct, BB_upper, SMA_50, SMA_20, volume_lag3
INFY	45	BB_pct, MACD_hist, BB_lower, MACD, MACD_signal, volume_z, volume_lag1, volume_lag3
MM	45	BB_lower, MACD, volume_z, volume_lag1, SMA_50, BB_pct, BB_upper, MACD_signal
RELIANCE	45	volume_lag1, volume_lag3, MACD_signal, BB_upper, SMA_20, volume_z, MACD, SMA_50

Common features across all stocks (37): date, returns, sector returns, macro variables, fundamentals, technical indicators, lagged returns, volume metrics, sentiment score.

3.3.3 Common Feature Selection Methods (Used in Both Approaches)

For either approach, let X_{all} be the final feature matrix used for feature selection and y_{res} be the residualized return target. Missing values are imputed with zeros. A standardized version of X_{all} is additionally created for scale-sensitive methods.

1) Domain-based Mutual Information (MI) Mutual information scores are computed between each feature and y_{res} to capture non-linear dependence. Features are grouped by domain (momentum, MACD, moving averages, Bollinger features, volume, lags, macro, FX, commodities, fundamentals, etc.), and only the highest-MI feature in each domain is retained.

2) Random Forest Importance A `RandomForestRegressor` is trained on X_{all}. Features are ranked by tree-based importance, and the smallest set explaining at least 90% cumulative importance is selected (with a minimum-feature fallback to avoid over-pruning).

3) RFE with Ridge Recursive Feature Elimination (RFE) is applied using a Ridge regressor on the standardized feature matrix. Features are iteratively removed until a fixed number of features (20 in this implementation) remains, providing a linear-model-based ranking.

4) SHAP-based Selection (LightGBM) A LightGBM regressor is trained and SHAP values are computed (if SHAP is available). Features are ranked by mean absolute SHAP contribution, and the top 20 are selected. If SHAP is unavailable, the implementation falls back to MI-based selection.

Majority-vote aggregation The final feature set is obtained by aggregating selections from the four methods. Each feature receives one vote per method if selected, and features with at least two votes are retained (`SELECTED_BY_MAJORITY`) for downstream modeling.

This design combines domain interpretability, non-linear relevance, linear ranking stability, and model-specific importance while reducing dependence on any single selection method.

3.4 Model Training, Selection, and Evaluation

To evaluate predictive robustness, we trained multiple tree-based ensemble models under both regression and classification formulations. Regression models predict the continuous residualized return, while classifiers predict the direction of the residualized return.

All models were tuned via grid search over predefined hyperparameter ranges using a **70:15:15 train–validation–test split**. Model selection was based on validation performance (RMSE for regressors and log-loss for classifiers). Final results are only on the strictly unseen test set.

For boosting models (LightGBM, XGBoost, and CatBoost), the number of boosting rounds/iterations was fixed at 1000, with early stopping (**30 rounds**), applied based on validation perfomance

Model	Task	Target	Hyperparameters Tuned	Selection Criterion
LightGBM	R	RR	learning_rate, max_depth, num_leaves, min_child_samples	Min validation RMSE
LightGBM	C	RD	learning_rate, max_depth, num_leaves, min_child_samples	Min validation LogLoss
XGBoost	R	RR	learning_rate, max_depth, subsample, colsample_bytree	Min validation RMSE
XGBoost	C	RD	learning_rate, max_depth, subsample, colsample_bytree	Min validation LogLoss
Random Forest	R	RR	n_estimators, max_depth, min_samples_leaf	Min validation RMSE
Random Forest	C	RD	n_estimators, max_depth, min_samples_leaf	Min validation LogLoss
CatBoost	R	RR	depth, learning_rate, l2_leaf_reg	Min validation RMSE

Table 4: Summary of trained models, prediction tasks, targets, tuned hyperparameters, and validation-based selection criteria.

NOTE:
RD: Residualized direction ; **RR**: Raw residualized return
R: Regression ; **C**: Classifier

3.5 Results and Evaluation

Performance was computed separately for regression and classification models across all 6 datasets. The **Emboldened** Model in for each stock is the best performing model

Metric	Purpose
R^2	Explained variance
MAE / RMSE	Error magnitude
Directional Accuracy	Sign prediction correctness
Accuracy	Classification correctness
High-Confidence Accuracy	Accuracy on top-confidence predictions

Dataset	Model	MAE	R^2
Infosys	LGBM	0.00889	0.0066
	XGB	0.00858	0.0532
	RF	0.00874	0.0127
	CatBoost	0.00871	0.0460
HUL	LGBM	0.00839	-0.0343
	XGB	0.00828	-0.0256
	RF	0.00840	-0.0262
	CatBoost	0.00819	-0.0008
HDFC	LGBM	0.00730	-0.0056
	XGB	0.00732	0.0017
	RF	0.00772	-0.2940
	CatBoost	0.00728	0.0111
Airtel	LGBM	0.01177	-0.0102
	XGB	0.01182	-0.0214
	RF	0.01208	-0.0846
	CatBoost	0.01146	0.0305
Mahindra	LGBM	0.01190	0.0040
	XGB	0.01190	0.0096
	RF	0.01203	-0.0123
	CatBoost	0.01185	0.0139
Reliance	LGBM	0.00864	0.0013
	XGB	0.00854	0.0155
	RF	0.00875	0.0049
	CatBoost	0.00859	0.0193

(a) Regressors

Dataset	Model	Acc	HC Acc
Infosys	LGBM	40.0%	20.0%
	XGB	42.0%	30.0%
	RF	54.0%	40.0%
HUL	LGBM	50.0%	51.2%
	XGB	52.8%	55.8%
	RF	56.1%	57.8%
HDFC	LGBM	47.9%	44.7%
	XGB	47.9%	42.1%
	RF	48.4%	30.0%
Airtel	LGBM	59.4%	71.4%
	XGB	60.4%	57.1%
	RF	64.4%	61.9%
Mahindra	LGBM	48.8%	58.1%
	XGB	49.8%	48.8%
	RF	54.0%	60.0%
Reliance	LGBM	49.8%	47.2%
	XGB	51.2%	54.0%
	RF	50.7%	60.5%

(b) Classifiers

Table 5: Test performance comparison across for the First Approach of Feature selection

Dataset	Model	MAE	R^2
Infosys	LGBM	0.01123	0.0313
	XGB	0.01138	-0.0270
	RF	0.01121	-0.0050
	CatBoost	0.01124	-0.0079
Mahindra	LGBM	0.01195	0.0090
	XGB	0.01207	-0.0008
	RF	0.01209	-0.0289
	CatBoost	0.01186	0.0093
Reliance	LGBM	0.00867	0.0019
	XGB	0.00870	0.0083
	RF	0.00972	-0.1472
	CatBoost	0.00869	0.0164
HDFC	LGBM	0.00711	-0.0097
	XGB	0.00717	-0.0327
	RF	0.00822	-0.3686
	CatBoost	0.00706	-0.0006
Airtel	LGBM	0.00957	-0.0375
	XGB	0.00945	-0.0102
	RF	0.00958	-0.0222
	CatBoost	0.00949	-0.0105

(a) Regressors — Method 2

Dataset	Model	Acc	HC Acc
Infosys	LGBM	53.2%	56.8%
	XGB	56.5%	68.2%
	RF	57.4%	68.2%
Mahindra	LGBM	50.7%	41.9%
	XGB	53.5%	44.2%
	RF	53.5%	62.8%
Reliance	LGBM	53.5%	44.2%
	XGB	54.0%	46.5%
	RF	52.6%	46.5%
HDFC	**LGBM**	50.5%	49.5%
	XGB	50.5%	48.1%
	RF	47.7%	37.8%
Airtel	LGBM	52.8%	45.5%
	XGB	53.7%	45.5%
	RF	49.1%	52.3%

(b) Classifiers — Method 2

Table 6: Test performance comparison across datasets for Method 2 feature construction.

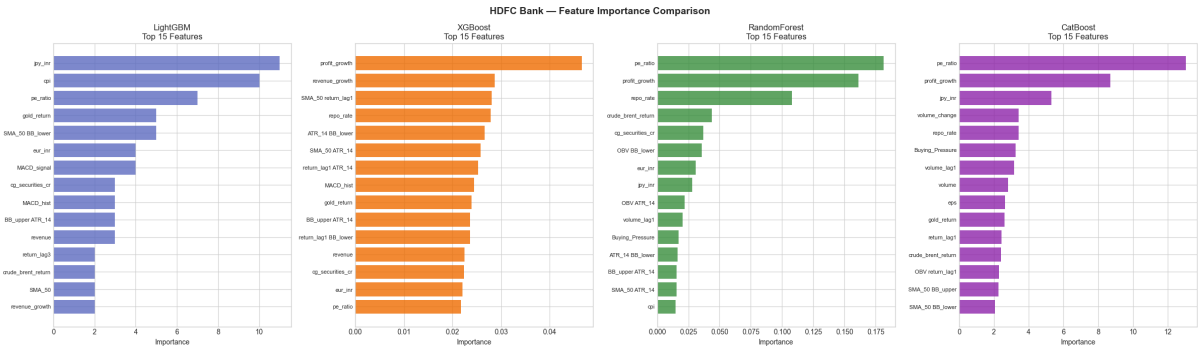


Figure 1: Sample Feature Weight Importance for Bharathi Airtel

4 Portfolio Design and Performance Evaluation

Portfolios were generated using Monte Carlo and mean–variance optimization, where the tangency portfolio achieved the maximum Information Ratio on the efficient frontier.

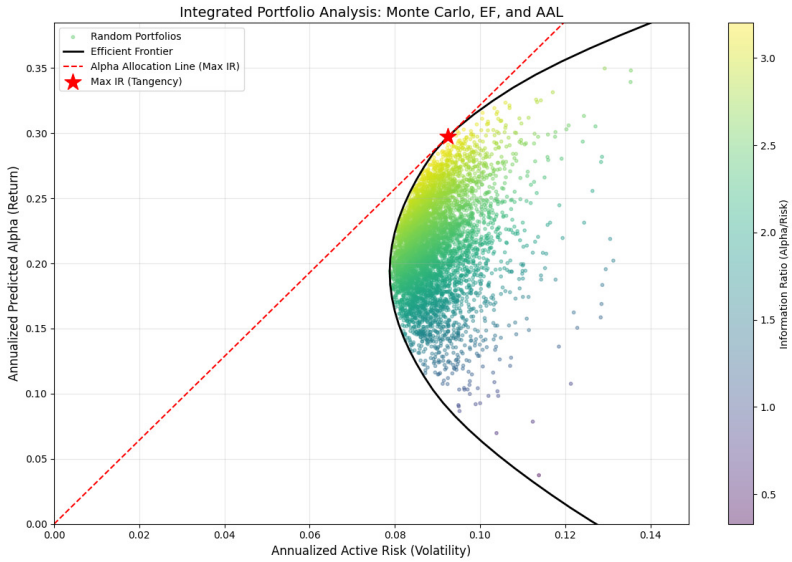
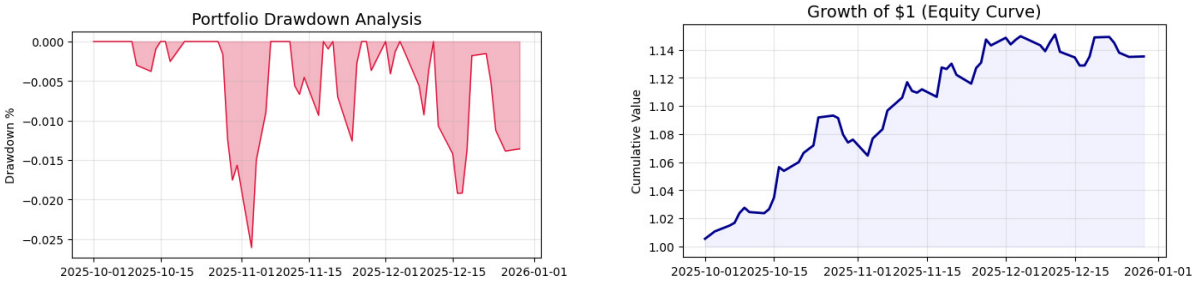


Figure 2: Efficient Frontier Portfolio Design

The wealth curve rises steadily with small drawdowns, and weights stay diversified and near optimal, indicating stable, robust performance.



(a) Portfolio Drawdown Curve (b) Equity Curve: Portfolio Growth

Figure 3: Portfolio Performance Analysis

Stock	Optimal Tangency (%)	Avg Tangency (%)
INFY	30.17	22.02
BHARTIARTL	25.82	20.84
HDFCBANK	12.44	21.89
MM	13.23	14.49
RELIANCE	18.35	13.23
HUL	0.00	7.53

Metric	Value
Annualized Return	56.86%
Annualized Volatility	11.57%
Information Ratio	4.9146
Avg. Sharpe Ratio	2.718
Max Drawdown	-4.34%
Directional Hit Ratio	59.68%

Table 7: Portfolio Weights and Backtest Performance Summary (Oct–Dec 2025)

5 Inferences

- The regression models show limited but non-zero predictive power overall, with only a few stocks/models achieving clearly positive R^2. This indicates that daily return prediction is difficult and noisy.
- Boosting-based models (especially XGBoost/CatBoost) perform more consistently than Random Forest in regression across the six stocks.
- Directional classification results are generally stronger than regression results, suggesting that predicting the *direction* of returns is easier than predicting the exact return value.
- Some stocks (such as HDFC in our results) remain hard to predict across both regression and classification, indicating weaker signal in the current feature set.
- Feature importance and selected features suggest that predictive information comes from a combination of technical indicators, lagged variables, macro variables, and selected fundamentals rather than any single feature group.
- Portfolio-level performance is better than what stock-level regression metrics alone suggest, as seen from the efficient frontier and cumulative return plots.
- The final portfolio weights are concentrated in a few stronger stocks but still remain reasonably diversified across the universe.

6 Conclusions

- The proposed multi-source forecasting pipeline is able to extract useful signals, but the overall predictive strength is moderate, which is expected for daily equity return forecasting.
- Classification-based predictions are more practically reliable than regression-based predictions in this assignment setup.
- Boosting models are generally the preferred choice for this task due to their better consistency across stocks.
- Even when stock-level prediction metrics are modest, the portfolio construction step helps translate signals into better economic outcomes.
- Overall, the framework is suitable as a baseline for multi-asset forecasting and portfolio allocation, with scope for improvement through better features and model tuning.

A Appendix

A.1 Raw Feature Description

Feature	Description
Open / High / Low / Close	Daily price levels capturing intraday valuation dynamics.
Volume	Shares traded, proxy for liquidity and participation.
Stock Log Return	Daily log return measuring price momentum.
Volume Change	Day-over-day change in trading activity.
EPS	Earnings per share indicating firm profitability.
PE Ratio	Valuation metric relative to earnings.
Revenue	Top-line business performance indicator.
Revenue Growth	Growth rate of company revenues.
Net Income	Bottom-line profitability measure.
Profit Margin	Operational efficiency ratio.
Profit Growth	Growth in net profitability.
NIFTY50 Return	Broad market return factor.
Sector Return	Industry-level return spillover effect.
India VIX	Market-implied volatility index.
Repo Rate	Central bank policy rate indicator.
91-Day T-Bill	Short-term risk-free yield proxy.
10Y G-Sec Yield	Long-term sovereign yield benchmark.
Yield Curve Spread	Term structure slope indicator.
CPI	Inflation measure.
CG / SG Securities Credit	Government borrowing and liquidity proxy.
FII / DII Net (Cr)	Institutional capital flow indicators.
USD-INR / EUR-INR / JPY-INR	Exchange rate exposure variables.
Crude Brent Return	Energy price sensitivity factor.
Gold Return	Global risk-aversion proxy.
Sentiment Score	Text-derived market sentiment indicator.
Target Raw Return	Forward stock return prediction target.
Target Residualized	Idiosyncratic return after factor adjustment.

Table A.1: Raw Feature Set Used for Return Prediction

A.2 Engineered Technical Indicators

Indicator	Category	Description
SMA (20, 50)	Trend	Short- and medium-term trend averages.
EMA (12, 26)	Trend	Recency-weighted moving averages.
MACD / Signal / Hist	Momentum	EMA spread and crossover momentum signal.
RSI (14)	Momentum	Overbought / oversold oscillator.
Bollinger Bands + BB_pct	Volatility	Dynamic $\pm 2\sigma$ price range indicator.
ATR (14)	Volatility	Range-based volatility measure.
OBV	Volume	Cumulative volume flow signal.
Volume Z-score	Volume	Standardized abnormal trading activity.
Buying / Selling Pressure	Price Action	Close location within daily high–low range.
Lagged Returns (1,3)	Momentum	Short-horizon return persistence.
Lagged Volume (1,3)	Volume	Short-horizon volume persistence.
Lagged Sector Return (1)	Market	Recent sector spillover effect.

Table A.2: Engineered Technical Indicators

A.3 Example Plots of Various Stocks

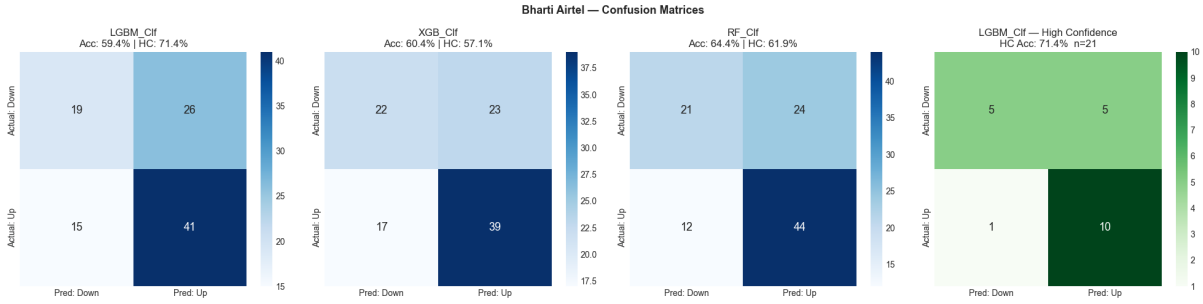


Figure 4: Confusion Matrix

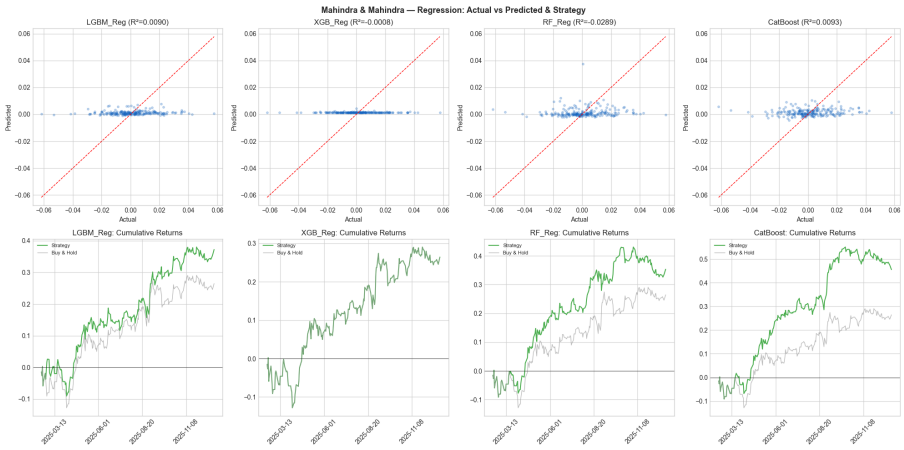


Figure 5: Actual vs Predicted Strategy

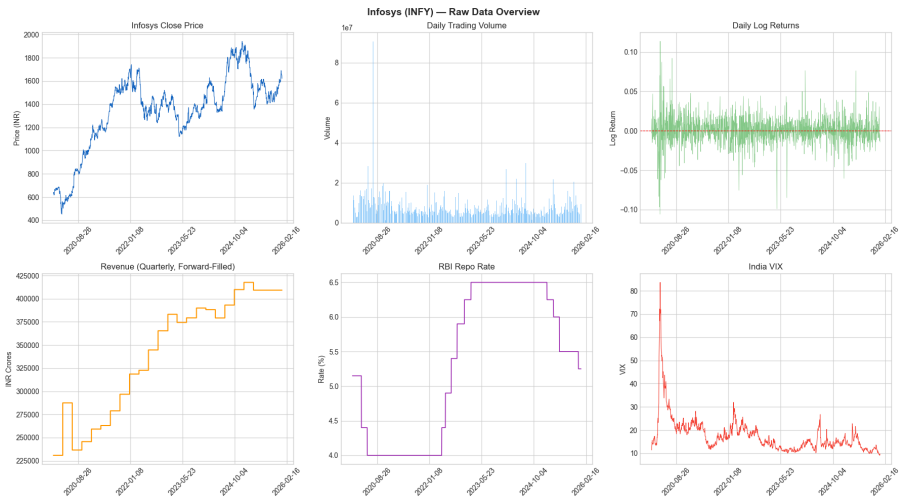


Figure 6: Sample Raw Data Overview